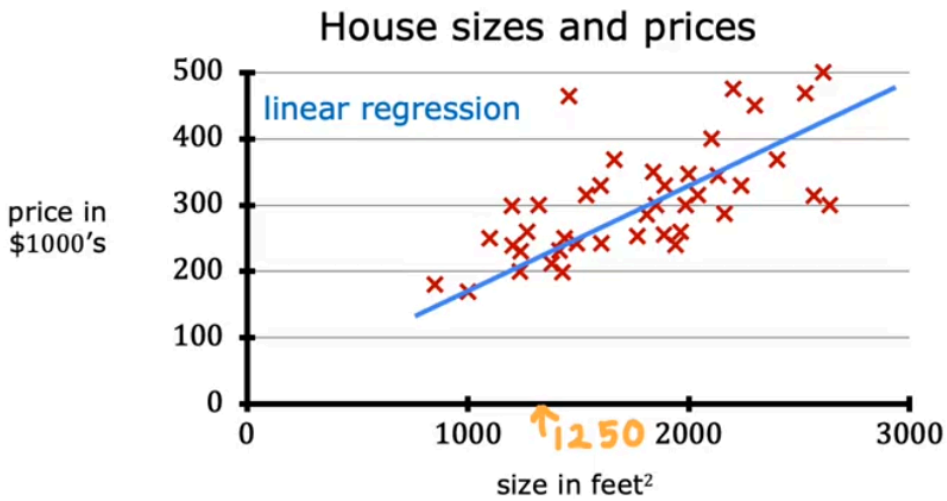
Day 2. Regression model

Linera regression



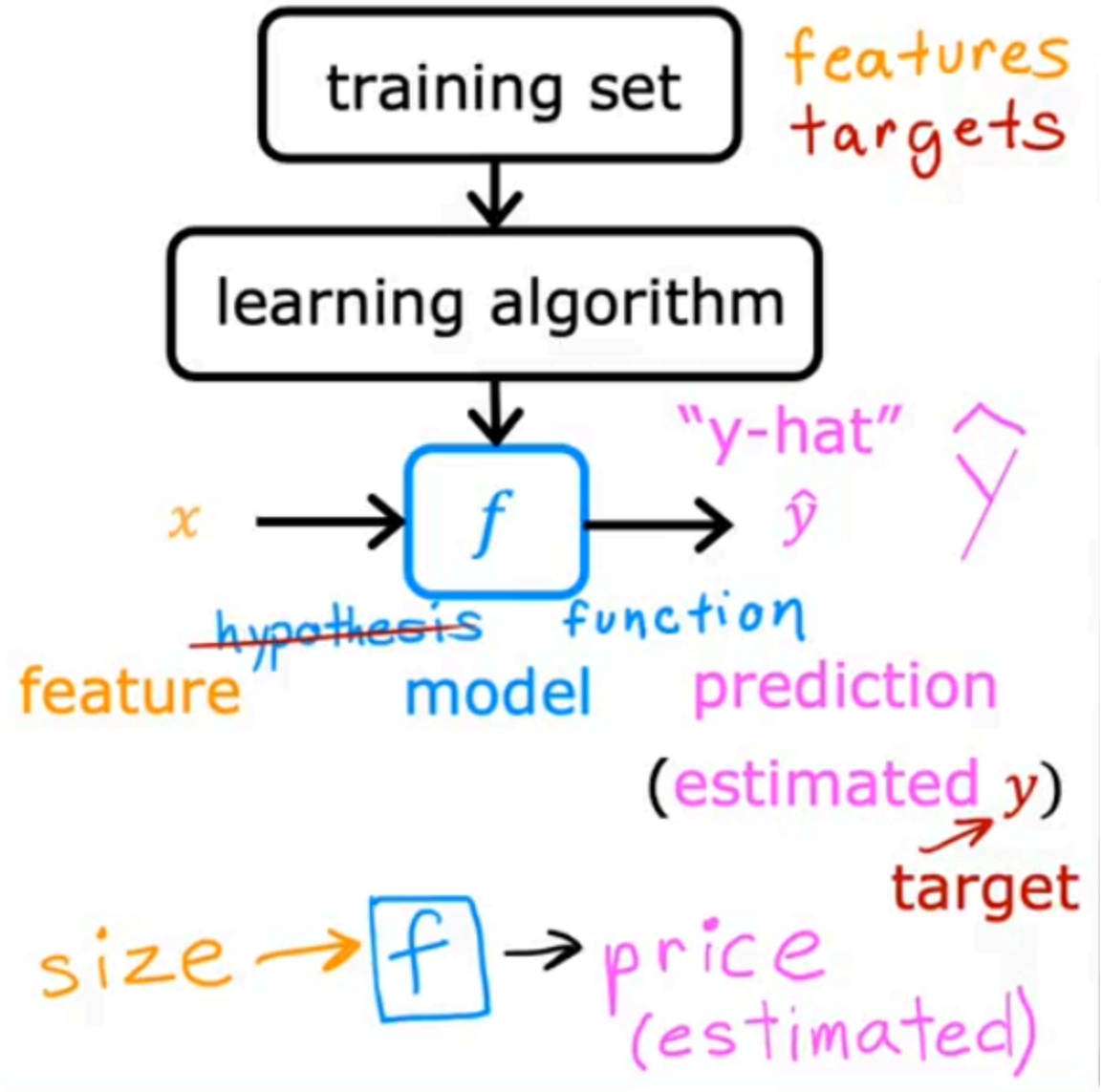
any supervised model that predicts number outputs are called regression model.

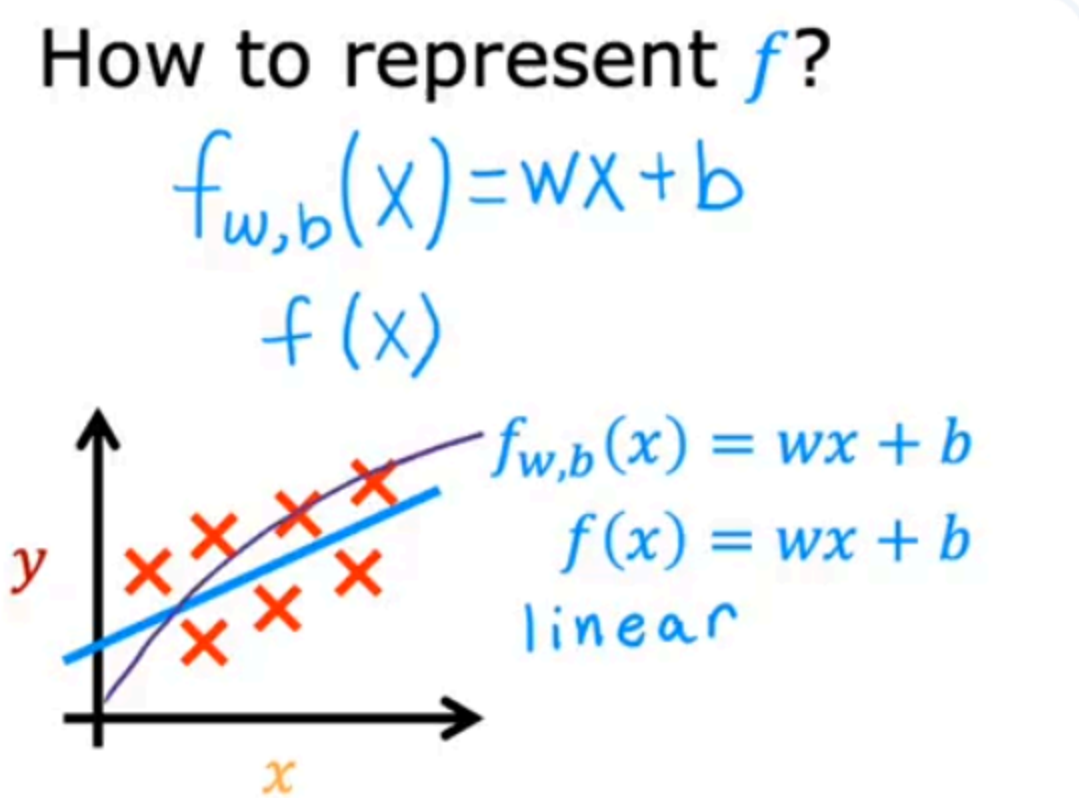
Terminology

- training set: data used to train the model
- x = "input" variable/feature
- y = "output"/"target" variable
- m = numberof training examples
- (x, y) = single training example
- $(x^{(i)}, y^{(i)}) = i^{th}$ training example

Process

what exactly does supervised learning do and how?





we will start from linear function, and we will learn non-linear functions later.

as for now, what we are going to do is to construct the function f(x) = wx + b, where w stands for weight and b stands for bias.

here the line is linear regression with one variable, we call it as:
univariate linear regression.

> - Linear regression builds a model which establishes a relationship between features and targets

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- In the example above, the feature was house size and the target was house
price
- for simple linear regression, the model has two parameters w and b whose
values are 'fit' using _training data_.
- once a model's parameters have been determined, the model can be used to
make predictions on novel data.
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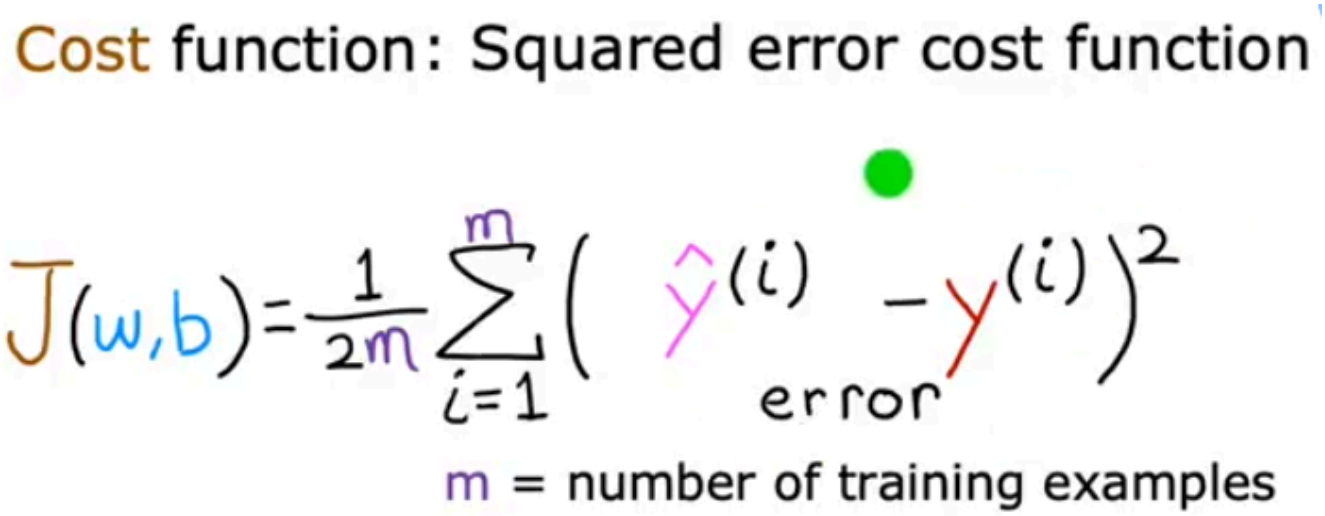
Cost functions

f(x) = wx + b
how do we determine the right w and b here? how do we find the w and b to make y_hat closer to $y^{(i)}$?

to do that, we construct a cost function.



in ML, people will use different loss function according to different situations, but here the square function is the most common one in regression.